

Question 1

Steps Heuristic

Let s be current state, g be goal state, n be length of both

$$\text{Formula: } h(s) = \sum_{i=1}^n \begin{cases} 1 & \text{if } s_i \neq g_i \\ 0 & \text{if } s_i = g_i \end{cases}$$

Since every action cost 1 in steps mode and every incorrect letter requires at least 1 action

It will be $h(s) \leq \text{optimal remaining number of steps}$

It will never overestimate as the heuristic relaxes the restraints and assumes it can immediately change a letter to the correct one.

For example, why to ask all three positions differ and thus it assumes only 3 actions are needed giving a estimated cost of 3 despite the optimal cost being 5. Even for dog to cat which can change straight to cat using the chain dog cog cot cat, it won't overestimate as each letter needs at least one action if wrong.

Scrabble Heuristic

Let s be current state, g be goal state, n be length of both, and $w(x)$ be the scrabble cost of a letter.

$$\text{Formula: } h(s) = \sum_{i=1}^n \begin{cases} w(g_i) & \text{if } s_i \neq g_i \\ 0 & \text{if } s_i = g_i \end{cases}$$

This heuristic is also relaxing the restraints to assume that it can immediately change a letter to the correct one. Therefore, it will be $h(s) \leq \text{optimal remaining scrabble cost}$

It also cannot overestimate as each letter that differs needs at least one action which will be the action of changing the letter to the goal state's letter which will be at minimum $w(g_i)$ it may require more actions but because scrabble costs contains no negative numbers it will never be lower than that minimum cost.

Frequency Heuristic

This Heuristic is basically just steps heuristic as the cost formula is $1 + \text{rarity}(w)$, therefore each action will cost at minimum 1.

$$\text{Formula: } h(s) = \sum_{i=1}^n \begin{cases} 1 & \text{if } s_i \neq g_i \\ 0 & \text{if } s_i = g_i \end{cases}$$

For the same reasoning of steps heuristic. Each letter that differs must take at least 1 action and that cost is $1 + \text{rarity}(w)$ with $\text{rarity}(w)$ being non-negative it will never take less than 1.

Question 2

	A	B	C	D	E	F	G
1	Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction
2	WHY → ASK	Steps	5	773	5	16	97.9301423
3	WHY → ASK	Scrabble	9	765	9	13	98.30065359
4	WHY → ASK	Frequency	38.86	808	38.86	755	6.559405941
5	BEAR → DUCK	Steps	4	691	4	6	99.1316932
6	BEAR → DUCK	Scrabble	11	2981	11	6	99.79872526
7	BEAR → DUCK	Frequency	38.24	1084	38.24	720	33.57933579

1. Yes.

2. why to ask, A* made the biggest difference for Scrabble.

bear to duck, A* made the biggest difference for Scrabble again.

3. Least informative for Frequency in both.

4. For heuristic “steps” it only counts number of incorrect positions, thus assumes each letter will take only one action. “Scrabble” uses the scrabble costs of the letters giving it a more accurate representation of the estimated cost. “Frequency” uses only the incorrect positions without making use of the $rarity(w)$ part of the formula thus it’s estimated cost doesn’t represent the actual cost nearly as much resulting in it being the least informative for A^* .

Question 3

Steps

1. Yes
2. Since the reverse chain chains the same amount of actions taken the cost will be the same as each action always cost 1
3. Not applicable answer to 1 was yes

Scrabble

1. No
2. Since the chain is in reverse the letter cost at the end will change.
3. For example. test to bush: test best bust bush.

test to bush

test to best cost 3

best to bust cost 1

bust to bush cost 1

Total Cost: 5

bush to test

bush to bust cost 1

bust to best cost 1

best to test cost 1

Total Cost: 3

As you can see the chain result in a different total value due to the letter being exchanged is different. With test to bush the letter changes are: b, u, h. While bush to test is: t, e, t.

Thus if the values for the letter differ than the cost will also differ.

Frequency

1. No

2. Since the chain is in reverse the word cost at the end will change.

3. For example. Dog to Cat: dog cog cot cat.

dog to cat

dog to cog cost 12.393166

cog to cot cost 11.071238

cot to cat cost 7.886906

Total Cost: 31.35131

cat to dog

cat to cot cost 11.071238

cot to cog cost 12.393166

cog to dog cost 6.819885

Total Cost: 30.284289

As you can see the difference occurs in the final step in which for dog to cat the final word cost is cat which is 7.886906. While cat to dog final word cost is dog which is 6.819885

Question 4

Model Used: ChatGPT

Prompt Given and its proposed Heuristic Function.

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary. The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

A clean rarity-aware heuristic is to use the **minimum rarity of any dictionary word** as a per-step lower bound.

Let

$$d(n, g) = \#\{i: n_i \neq g_i\}$$

be the Hamming distance from state n to the goal g , and let

$$r_{\min} = \min_{w \in D} \text{rarity}(w),$$

where D is the supplied dictionary.

Define

$$h(n) = \begin{cases} 0, & n = g, \\ d(n, g) (1 + r_{\min}), & n \neq g. \end{cases}$$

3. This heuristic is admissible because it determines the amount of positions that differ in initial and goal which is the hamming distance and multiplies that by $(1 + r_{\min})$. For example, dog to cat's hamming distance is 3 and all 3 positions differ. $r_{\min}$ is the value of the word with the

lowest frequency value in the dictionary. Thus $1 + \text{rmin}$ will always be equal to or less than $1 + \text{rarity}(w)$ which is the actual cost formula. It is also assuming that it can reach the goal state by changing each letter once which will result in equal to or less than the optimal amount of actions needed.

4. Not applicable the heuristic is admissible.

5.

8	ChatGPT Heuristic	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction
9	WHY → ASK	Frequency	38.86	808	38.86	754	6.683168317
10	BEAR → DUCK	Frequency	38.24	1084	38.24	714	34.13284133

Solution Costs are still the same optimal cost and slight improvement in A* nodes expanded.

Thus, the evidence supports using its heuristic.